

	d) <i>Mean:</i> $\mu X + 4Y = \mu X + 4\mu Y = 9.5 + 4 \times 6.8 = 9.5 + 27.2 = 36.7$ <i>Std. deviation:</i> $\sigma X + 4Y = \sqrt{(\sigma X)^2 + (4\sigma Y)^2} =$ $\sqrt{(0.4)^2 + (4 \times 0.1)^2} = \sqrt{0.16 + 0.16} = \sqrt{0.32} \approx 0.5657$	
13.	A laminated item is composed of five layers. The layers are a simple random sample from a population whose thickness has mean 1.2 mm and standard deviation 0.04 mm. a. Find the mean thickness of an item. b. Find the standard deviation of the thickness of an item. Ans: Given: Each layer's thickness (X) is a random sample from a population Mean (μ_X) = 1.2 mm Standard deviation (σ_X) = 0.04 mm Total thickness (S) = $X_1 + X_2 + X_3 + X_4 + X_5$ Each X_i is independent and identically distributed Mean thickness of the item (mean of sum = sum of means) $\mu_S = 5 \times \mu_X = 5 \times 1.2 = 6.0 \text{ mm}$ Std. deviation of the thickness of the item (for independent layers, variance of sum = sum of variances) $Var(S) = \sum_{i=1}^5 Var(X_i) = 5 \times \sigma_X^2 = 5 \times (0.04)^2 = 5 \times 0.0016 = 0.008$ $\sigma_S = \sqrt{Var(S)} = \sqrt{0.008} \approx 0.0894 \text{ mm}$	Exercise 2.5
14.	Suppose that we extract an individual at random from a population whose members have an average income of \$40,000, with a standard deviation of \$20,000. What is the probability of extracting an individual whose income is either less than \$10,000, or greater than \$70,000? Ans: Mean income $\mu = \\$40,000$ Std. deviation $\sigma = \\$20,000$ We want the probability $P(X < \\$10,000 \text{ or } X > \\$70,000)$ Distance from the mean: $\\$10,000 - \\$40,000 = \\$70,000 - \\$40,000 = \\$30,000$ in terms of std deviation $\frac{\\$30,000}{\\$20,000} = 1.5\sigma$ So, we are looking for the probability that X is at least 1.5 standard deviations away from the mean: $P(X - \mu \geq 1.5\sigma)$	Chebyshev's inequality

	Chebyshev's inequality states: $P(X - \mu \geq k\sigma) \leq \frac{1}{k^2}$ $P(X - \mu \geq 1.5\sigma) \leq \frac{1}{1.5^2} = \frac{1}{2.25} \approx 0.444$ The probability that an individual's income is less than \$10,000 or greater than \$70,000 is at most 44.4%	
15.	Suppose the average number of requests a web server receives per minute is 100, with a standard deviation of 10. What is the minimum percentage of time the number of requests will be between 80 and 120 Ans: Mean = 100, std deviation = 10, Range: 80-120 Number of std deviations from the mean: $80 = 100 - 20 = 100 - 2 \times 10$ $120 = 100 + 20 = 100 + 2 \times 10$ So, 80 and 120 are both 2 standard deviations away from the mean Chebyshev's inequality states that at least $1 - \frac{1}{k^2}$ of the data lies within k standard deviations of the mean (for any distribution) Here, $k=2$ $1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$ At least 75% of the time, the number of requests will be between 80 and 120	Chebyshev's inequality
16.	A program has an expected execution time of 10 seconds with a standard deviation of 2 seconds. Using Chebyshev's inequality, what is the probability that the program will take longer than 16 seconds to execute? Ans: Mean execution time, $\mu = 10$ seconds Std deviation $\sigma = 2$ seconds We want $P(X > 16)$ Step 1: Express in terms of deviations from the mean $16 - 10 = 6$ seconds Calculate how many std. deviations above the means: $k = \frac{16 - 10}{2} = 3$ Step 2: Apply Chebyshev's inequality Since we want $P(X > 16)$ which is	Chebyshev's inequality

$$P(X - \mu \geq 3\sigma) \leq P(|X - \mu| \geq 3\sigma) \leq \frac{1}{3^2} = \frac{1}{9} \approx 0.1111$$

The probability that the program takes more than 16 seconds to execute is at most 11.11% (upper bound)

17.

As part of a quality-control study aimed at improving a production line, the weights (in ounces) of 20 bars of soap are measured. The results are as follows, sorted from smallest to largest.

11.6 12.6 12.7 12.8 13.1 13.3 13.6 13.7 13.8 14.1
14.3 14.3 14.6 14.8 15.1 15.2 15.6 15.6 15.7 15.8

Construct a normal probability plot for these data. Do these data appear to come from an approximately normal distribution?

Ans:

Calculate expected normal quantiles (theoretical quantiles):

- For each observation i in the sorted data ($i=1,2,\dots,20$), calculate the plotting position:

$$p_i = \frac{i - 0.5}{n} \text{ where } n = 20$$

Find the corresponding z-scores (quantiles of standard normal distribution) for these probabilities p_i

Plot the observed data values (y_i) against the theoretical z-scores (x_i)

If the data points fall approximately on a straight line, the data can be considered approximately normal.

Quick calculation example for first and last data points:

- For $i = 1$:

$$p_1 = \frac{1 - 0.5}{20} = \frac{0.5}{20} = 0.025$$

z-score for 0.025 $\approx$ -1.96

- For $i = 20$:

$$p_{20} = \frac{20 - 0.5}{20} = \frac{19.5}{20} = 0.975$$

z-score for 0.975 $\approx$ +1.96

If you plot these points and see an approximate straight line, the weights appear to be normally distributed.

Exercise
4.10

18.	Bottles filled by a certain machine are supposed to contain 12 oz of liquid. In fact the fill volume is random with mean 12.01 oz and standard deviation 0.2 oz. a. What is the probability that the mean volume of a random sample of 144 bottles is less than 12 oz? b. If the population mean fill volume is increased to 12.03 oz, what is the probability that the mean volume of a sample of size 144 will be less than 12 oz? Ans: a) Population mean $\mu = 12.01\text{oz}$ Population std deviation, $\sigma = 0.2\text{ oz}$ Sample size, $n = 144$ By CLT, the sampling distribution of the mean $\bar{X}$ is approximately normal with Mean: $\mu_{\bar{X}} = \mu$ Standard deviation (standard error): $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{0.2}{\sqrt{144}} = \frac{0.2}{12} = 0.01667$ Part (a): Probability mean volume of sample < 12 oz (with $\mu = 12.01$) Calculate the z-score: $z = \frac{12 - 12.01}{0.01667} = \frac{-0.01}{0.01667} \approx -0.6$ Using z-tables or normal distribution calculators, $P(\bar{X} < 12) = P(Z < -0.6) \approx 0.2743$ So, the probability is about 27.43%. Part (b): μ increases to 12.03 oz, find $P(\bar{X} < 12)$: $z = \frac{12 - 12.03}{0.01667} = \frac{-0.03}{0.01667} \approx -1.8$ Using z-tables: $P(\bar{X} < 12) = P(Z < -1.8) \approx 0.0359$ So, the probability is about 3.59%.	Exercise 4.11
19.	A 500-page book contains 250 sheets of paper. The thickness of the paper used to manufacture the book has mean 0.08 mm and standard deviation 0.01 mm. a. What is the probability that a randomly chosen book is more than 20.2 mm thick (not including the covers)? b. What is the 10th percentile of book thicknesses? c. Someone wants to know the probability that a randomly chosen page is more than 0.1 mm thick. Is enough information given to compute this probability? If so, compute the probability. If not, explain why not.	Exercise 4.11

Ans:

Number of sheets $n = 250$

Thickness of paper

$$\mu = 0.08\text{mm}, \sigma = 0.01\text{ mm}$$

a) Probability the book thickness is more than 20.2 mm

Total book thickness (excluding covers) is the sum of thicknesses of 250 sheets.

By the Central Limit Theorem:

Mean total thickness:

$$\mu_{total} = n \times \mu = 250 \times 0.08 = 20\text{ mm}$$

Standard deviation of total thickness:

$$\sigma_{total} = \sqrt{n} \times \sigma = \sqrt{250} \times 0.01 \approx 15.81 \times 0.01 = 0.1581\text{ mm}$$

Calculate the z-score for 20.2 mm:

$$z = \frac{20.2 - 20}{0.1581} \approx \frac{0.2}{0.1581} \approx 1.265$$

Using standard normal distribution tables,

$$P(X > 20.2) = 1 - P(Z \leq 1.265) \approx 1 - 0.8962 = 0.1038$$

The probability that the book is thicker than 20.2 mm is approximately 10.38%

b) Find the 10th percentile of book thicknesses

The 10th percentile corresponds to $z = -1.28$ (from standard normal tables)

Using the z-score formula:

$$X = \mu_{total} + z \times \sigma_{total} = 20 + (-1.28) \times 0.1581 \approx 20 - 0.202 = 19.798\text{ mm}$$

The 10th percentile is approximately 19.8mm

c) Probability a randomly chosen page is more than 0.1mm thick

We are asked for $P(\text{page thickness} > 0.1)$

Give the page thickness mean $\mu = 0.08\text{mm}$ and std deviation $\sigma = 0.01\text{mm}$

Assume the page thickness follows a normal distribution (usually reasonable for measurement data)

Calculate the z-score:

$$z = \frac{0.1 - 0.08}{0.01} = \frac{0.02}{0.01} = 2$$

Find $P(X > 0.1)$:

$$P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228$$

Yes, enough information is given. The probability that a randomly chosen page is thicker than 0.1 mm is approximately 2.28%.

20.

A commuter encounters four traffic lights each day on the way to work. Let X represent the number of these that are red lights. The probability mass function of X is as follows.

x	0	1	2	3	4
$P(X = x)$	0.1	0.3	0.3	0.2	0.1

What is the probability that in a period of 100 days, the average number of red lights encountered is more than 2 per day?

Ans:

Step 1: Find the mean of

X (expected number of red lights per day):

$$\begin{aligned}\mu &= \sum x \cdot P(X=x) = 0 \times 0.1 + 1 \times 0.3 + 2 \times 0.3 + 3 \times 0.2 + 4 \times 0.1 \\ &= 0 + 0.3 + 0.6 + 0.6 + 0.4 = 1.9\end{aligned}$$

Step 2: Find the variance of X :

First calculate $E(X^2)$: $\sum$

$$\begin{aligned}x^2 \cdot P(X = x) &= 0^2 \cdot 0.1 + 1^2 \cdot 0.3 + 2^2 \cdot 0.3 + 3^2 \cdot 0.2 + 4^2 \cdot 0.1 \\ &= 0 + 0.3 + 4 \times 0.3 + 9 \times 0.2 + 16 \times 0.1 = 0.3 + 1.2 + 1.8 + 1.6 = 4.9\end{aligned}$$

Variance:

$$\sigma^2 = E(X^2) - \mu^2 = 4.9 - (1.9)^2 = 4.9 - 3.61 = 1.29$$

Standard deviation:

$$\sigma = \sqrt{1.29} \approx 1.135$$

Step 3: Use CLT for sample mean over 100 days

Sample size $n = 100$

Exercise
4.11

Mean of sample mean:

$$\mu_{\bar{X}} = \mu = 1.9$$

Standard deviation of sample mean (standard error):

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1.135}{\sqrt{100}} = \frac{1.135}{10} = 0.1135$$

Step 4: Find the probability average number of red lights per day is more than 2

Calculate the z-score for $\bar{X} = 2$:

$$z = \frac{2 - 1.9}{0.1135} = \frac{0.1}{0.1135} \approx 0.88$$

Using standard normal tables or calculator:

$$P(\bar{X} > 2) = P(Z > 0.88) = 1 - P(Z \leq 0.88) \approx 1 - 0.8106 = 0.1894$$

The probability that the average number of red lights per day over 100 days is more than 2 is approximately 18.94%.

21.

The time taken by an automobile mechanic to complete an oil change is random, with mean 29.5 minutes and standard deviation 3 minutes.

- What is the probability that 50 oil changes take more than 1500 minutes?
- What is the probability that a mechanic can complete 80 or more oil changes in 40 hours?

Ans:

Calculate using CLT

- Mean time per oil change, $\mu = 29.5$ minutes
- Standard deviation, $\sigma = 3$ minutes

- Probability that 50 oil changes take more than 1500 minutes

Total time for 50 oil changes: $T = \sum_{i=1}^{50} X_i$

By CLT, T is approximately normally distributed with

Mean: $\mu_T = 50 \times 29.5 = 1475$ minutes

Std deviation: $\sigma_T = \sqrt{50} * 3 = 3 * 7.071 = 21.213$ minutes

Find the z-score:

$$z = \frac{1500 - 1475}{21.213} = \frac{25}{21.213} \approx 1.18$$

- Using standard normal table,

$$P(T > 1500) = P(Z > 1.18) = 1 - P(Z \leq 1.18) \approx 1 - 0.8810 = 0.1190$$

Probability is approximately 11.9%

Exercise
4.11

	b) Probability that a mechanic completes 80 or more oil changes in 40 hours 40 hrs = 2400 minutes Let N = no. of oil changes completed in 2400 minutes each oil change time X_i has mean 29.5, sd 3 We approximate N by considering the total time S_n for n oil changes $P(N \geq 80) = P(S_{80} \leq 2400)$ By CLT: Mean for 80 oil changes: $\mu_{S_{80}} = 80 \times 29.5 = 2360 \text{ minutes}$ Standard deviation: $\sigma_{S_{80}} = \sqrt{80} \times 3 = 3 \times 8.944 = 26.833 \text{ minutes}$ Calculate z-score: $z = \frac{2400 - 2360}{26.833} = \frac{40}{26.833} \approx 1.49$ Find probability: $P(S_{80} \leq 2400) = P(Z \leq 1.49) \approx 0.9319$ The probability that the mechanic completes 80 or more oil changes in 40 hours is approximately 93.19%	
22.	There are 100 men on a plane. Let X_i be the weight (in pounds) of the ii-th man, with X_i's being i.i.d., $E[X_i] = \mu = 170$, and $\sigma X_i = 30$. Find the probability that the total weight exceeds 18,000 pounds. Ans: Number of men, $n=100$ Each man's weight X_i is i.i.d with mean $\mu = 170$ pounds and std deviation $\sigma = 30$ pounds Total weight $S = \sum_{i=1}^{100} X_i$ Find $P(S > 18,000)$	CLT

	Step 1: Mean and standard deviation of total weight S $\mu_S = n \times \mu = 100 \times 170 = 17,000$ $\sigma_S = \sqrt{n} \times \sigma = \sqrt{100} \times 30 = 10 \times 30 = 300$ Step 2: Calculate z-score for $S = 18,000$ $z = \frac{18,000 - 17,000}{300} = \frac{1,000}{300} = 3.33$ Step 3: Use standard normal distribution to find $P(S > 18,000)$ $P(S > 18,000) = P(Z > 3.33)$ Using z-tables or calculator: $P(Z > 3.33) \approx 1 - 0.99957 = 0.00043$ The probability that the total weight exceeds 18,000 pounds is approximately 0.043% (very small).	
23.	Generating Poisson Variates Random Numbers: 0.25, 0.70, 0.15, 0.90, 0.40 Goal: Generate 3 Poisson variates ($\lambda = 0.2$) Ans: $\alpha = 0.2, e^{-0.2} = 0.8187$ Variate 1 $n=0, P=1$ $R_1 = 0.25, P = 1 * 0.25$ $0.25 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Variate 2 $n=0, P=1$ $R_2 = 0.70, P = 1 * 0.70$ $0.70 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Variate 3 $n=0, P=1$ $R_3 = 0.15, P = 1 * 0.15$ $0.15 < e^{-0.2} \Rightarrow \text{Accept } N=0$ Hence the three variates are 0,0 and 0.	Random variate generation